Infrastructure of the AI Era: A modular OCP system approach

HPC SYSTEMS

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HPC Systems



The AI factory has one output: tokens

The constraint has moved from silicon to infrastructure. Three measures decide what a site earns.

01 Time to first token

GPUs arrive faster than powered, cooled capacity. Idle racks earn nothing.

Modular answer: Factory-built modules deploy faster and more predictably.

02 Tokens per watt

Power is the scarce input. Conversion loss and cooling overhead are tokens not made.

Modular answer: 800V DC sidecar power and liquid cooling put more of each MW into compute.

03 Tokens over the lifetime

Every GPU generation raises rack density. Rigid infrastructure gets stranded.

Modular answer: Swap or add a power or cooling module, not the whole row.

Every Infrastructure constraint means fewer tokens

Illustrative figures for a 100 MW AI site

Deploy sooner. Put more power into compute. Adapt across technology generations

Idle

72 GWh

unused energy per month

for each month a finished 100 MW hall waits on power and cooling.

Modular answer: Factory-built, pre-tested modules shorten the path to energisation.

Wasted

+6 MW

additional IT capacity

from the same 100 MW grid connection by improving PUE from 1.3 to 1.2

Modular answer: 800V DC power and liquid cooling reduce conversion and cooling overhead.

Stranded

15+ yrs

facility investment horizon

Facility infrastructure lasts much longer than successive chip platforms and their changing power and cooling requirements

Modular answer: Upgrade power and cooling modules while retaining the rack and building interfaces.

Density is outrunning the infrastructure around it

<30 kW

Legacy air-cooled 19" racks

140 kW

48 VDC in-rack conversion, the practical ceiling

500+ kW

800 VDC distribution, sidecar power

~1 MW

Next-generation rack-scale systems

The problems are now coupled

Power Current, copper volume and conversion loss grow with every step. 48 V runs out of road.

Cooling Direct liquid cooling is mandatory; heat flux and hotspots shrink thermal margins.

Mechanics Payload, footprint and structural integrity become design limits, not afterthoughts.

Efficiency Scarce, expensive power means every 0.1 of PUE saved is roughly 8% more IT capacity.

Standardise the base. Modularise the rest.

Three independent modules on one standard base. Each scales on its own curve

IT

Customer's choice of rack-scale compute

Power

Sidecar 800V DC, RiLineX busbar, BBU

Cooling

In-rack & in-row CDU, RDHx, TCS

OCP rack foundation

ORV3 / ORW · common interfaces · certified · built at scale

Customers choose how to disaggregate

Today	Cooling moves to dedicated in-row CDUs.
Next	Power moves to a dedicated sidecar and 800 VDC.
Always	The rack interface stays the same.

Built on OCP work: Open Rack V3 · Deschutes · Diablo 400

The rack is now the platform



70%

projected OCP-based share of the market by 2030*

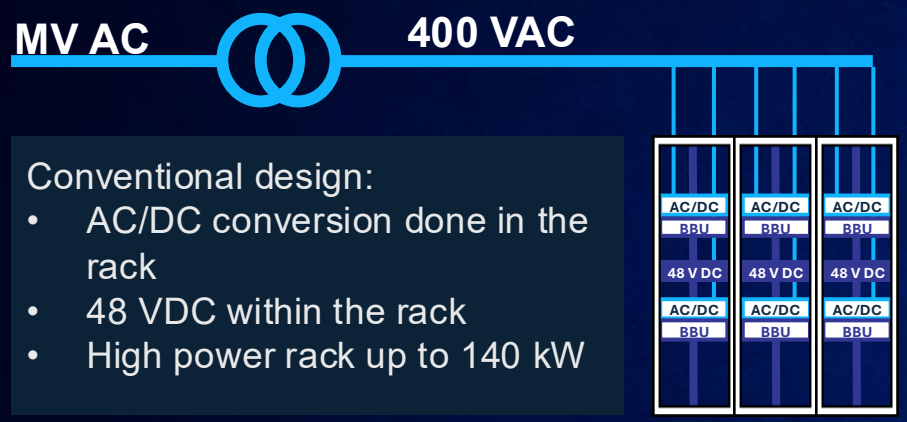
One family, three routes
ORV3-based, ORW and OCP spin-offs from a common mechanical architecture.

Room for liquid and power
21"/42" formats with space for manifolds, busbars and cabling.

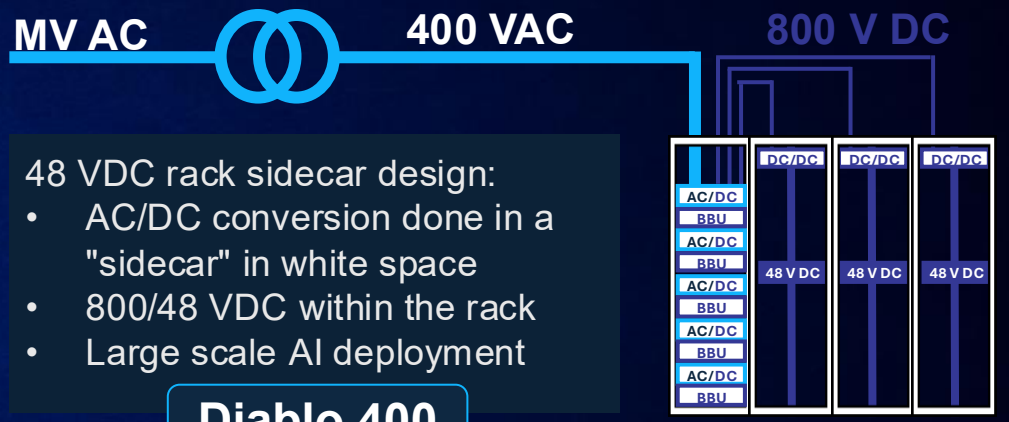
Qualify once, deploy globally
Common interfaces and UL certification, manufactured at scale.

*Source: Omdia 2025

Power: the path to 800 VDC

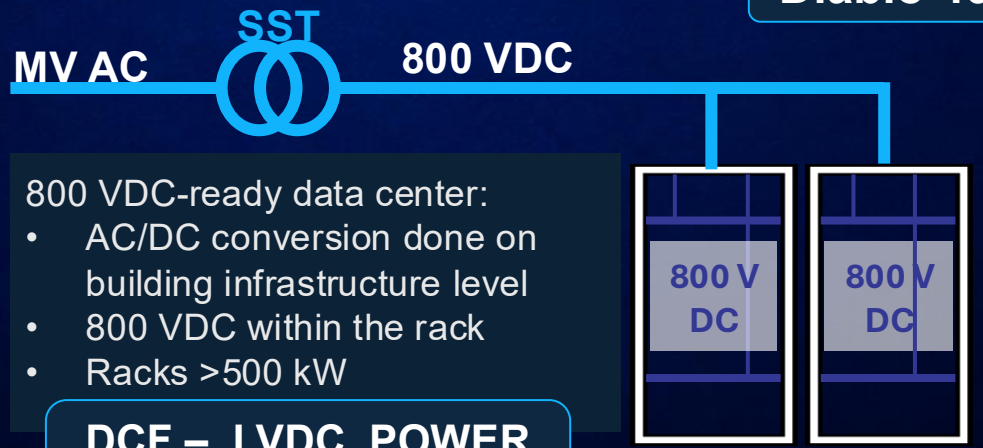


Conventional design:
- AC/DC conversion done in the rack
- 48 VDC within the rack
- High power rack up to 140 kW



48 VDC rack sidecar design:
- AC/DC conversion done in a "sidecar" in white space
- 800/48 VDC within the rack
- Large scale AI deployment

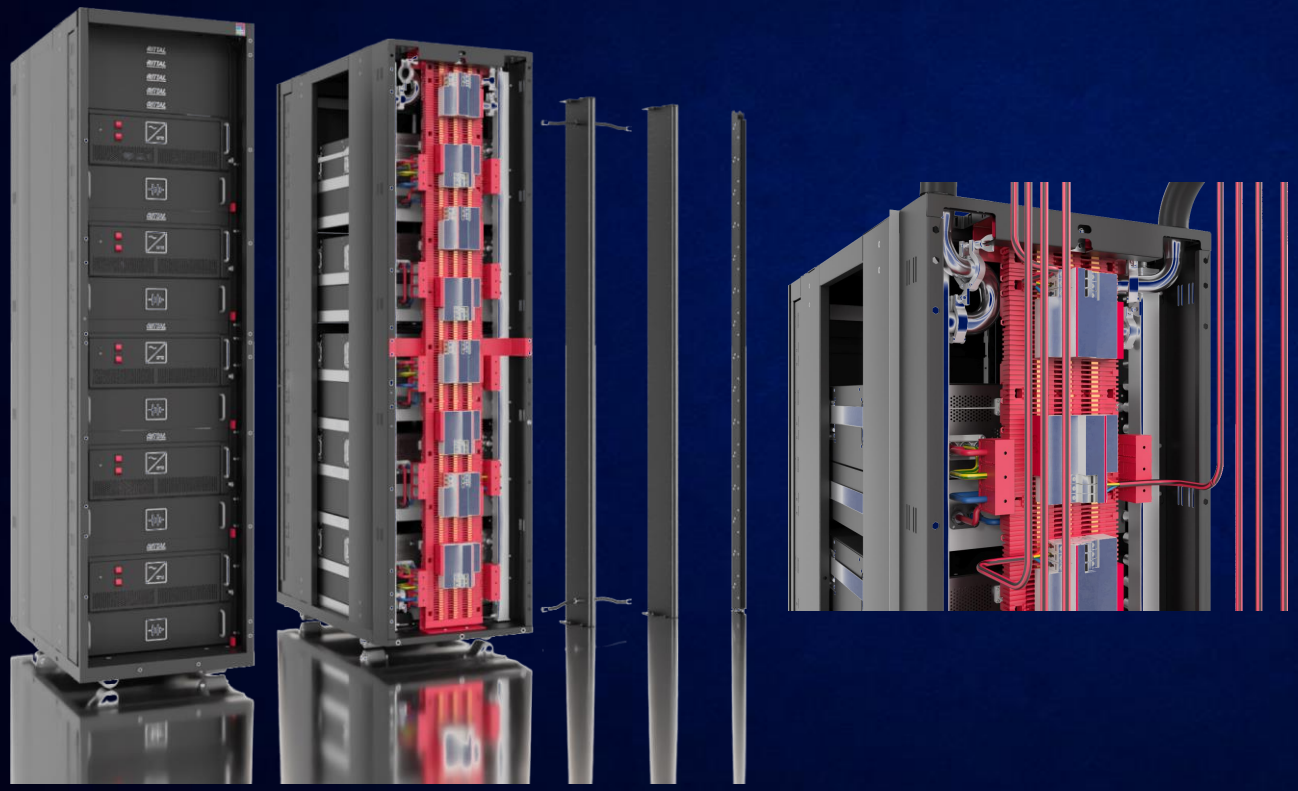
Diablo 400



800 VDC-ready data center:
- AC/DC conversion done on building infrastructure level
- 800 VDC within the rack
- Racks >500 kW

DCF – LVDC POWER DISTRIBUTION

Rittal power sidecar: the open platform for 800V DC



800 VDC
scalable distribution

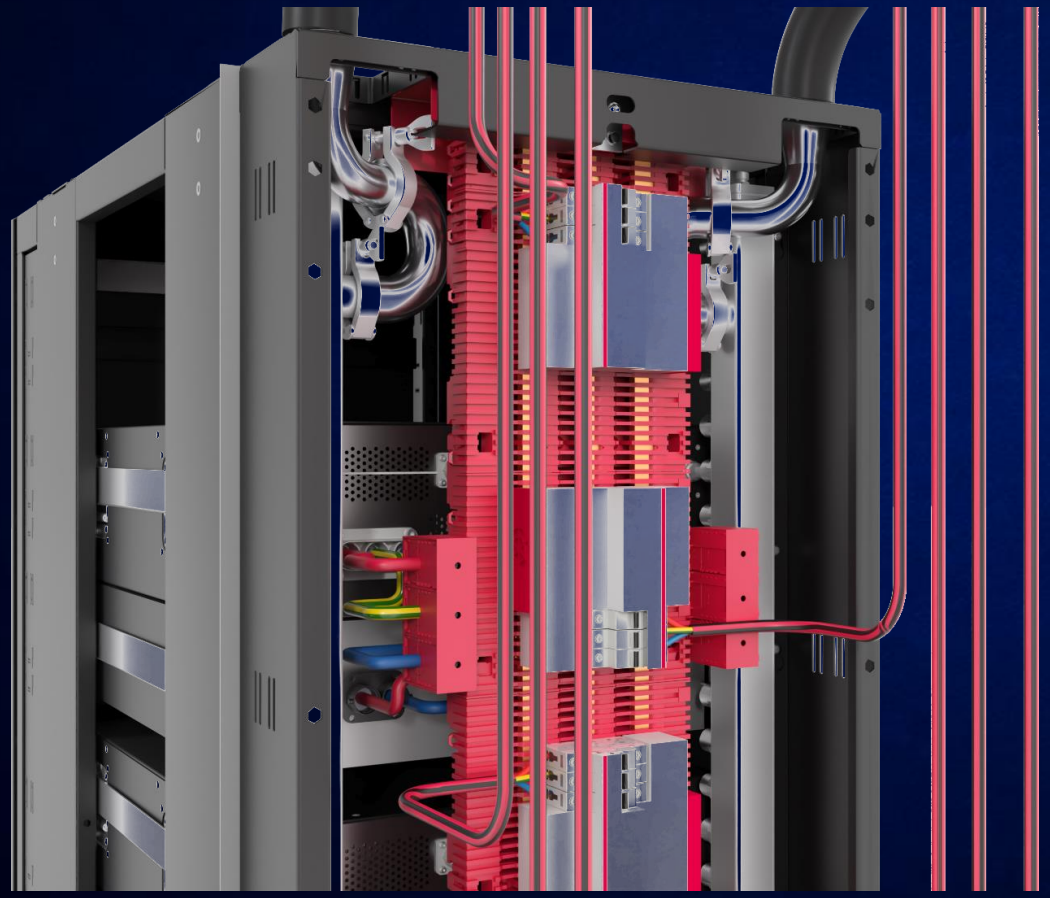
up to **2** MW
power rating per sidecar

Touch protected RiLineX backbone
Certified, touch-safe busbar system: safe, reliable, highly available.

Aligned with OCP Specifications
Open platform developed with technology partners and based on ORv3 Rack.

Serviceable by design
Power capacity added and maintained during live operation without touching the IT rack.

RiLineX: the open power backbone



1500 V

DC rating, 1000 V AC

65 kA

tested short-circuit resistance

Touch-safe by design

Integrated contact protection to IP 2XB, upgradable to IP 4X.

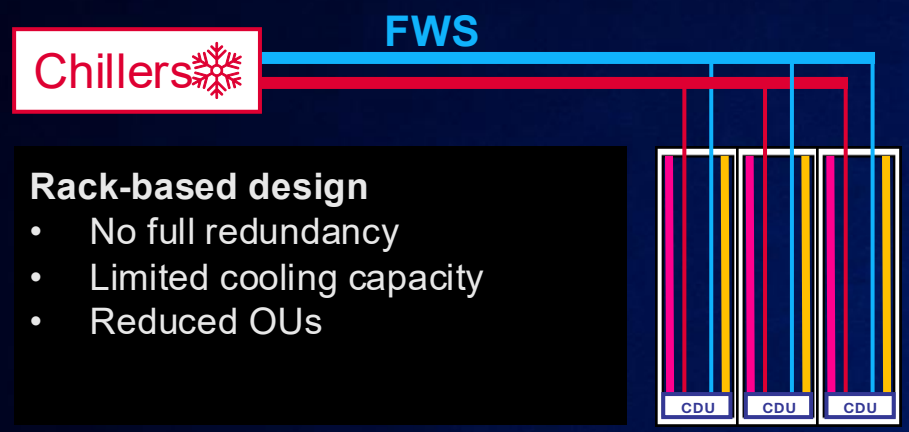
Open and licence-free

Manufacturer-neutral 60 mm platform with a technology-partner ecosystem.

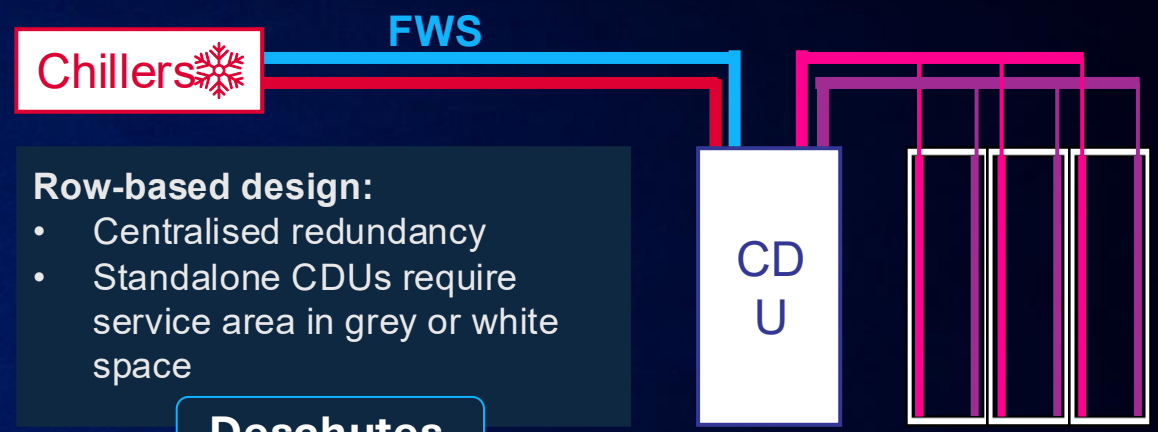
Faster to build, approved globally

Tool-free or pre-assembled boards; IEC 61439 and UL 508A approvals.

Cooling: from rack-based to in-row

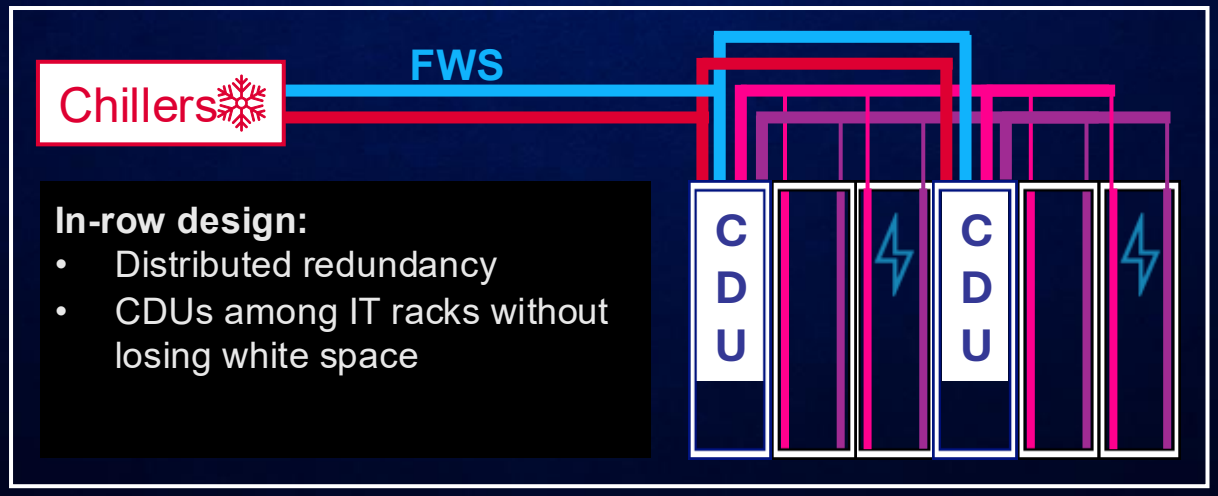


Rack-based design
- No full redundancy
- Limited cooling capacity
- Reduced OUs



Row-based design:
- Centralised redundancy
- Standalone CDUs require service area in grey or white space

Deschutes



In-row design:
- Distributed redundancy
- CDUs among IT racks without losing white space

Rittal liquid cooling: one OCP platform, every topology

80 kW

rear-door HEX,
48 VDC power

250 kW

in-rack CDU at
4K ATD

1.25MW

in-row CDU at
4K ATD

1.5MW grouped up to **13.5MW**

@ N+1





Same building blocks, from one CDU to a hall



Scale

In Row Stack 1500

1.5MW

to 13.5MW and beyond

Faster to first token factory-built, pre-validated modules

Tokens for life upgrade modules as density rises

Lower risk one standard, qualified once, deployed everywhere

Same building blocks, from one rack to a hall



Scale

250 kW

to 13.5MW…..then multiply

Faster to first token factory-built, pre-validated modules

Tokens for life upgrade modules as density rises

Lower risk one standard, qualified once, deployed everywhere

In summary

Standardise for scale.

Modularise for flexibility.

Keep every megawatt producing tokens.

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Thank you

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